

TASK-3

Title: Static Timing Analysis (STA) and Timing Closure of 8-Bit Multiplier Using Vivado

Objective: The objective of this task is to perform Static Timing Analysis (STA) on an 8-bit multiplier design and verify that the design meets the specified timing constraints after synthesis and implementation.

Tools Used: Verilog HDL, Xilinx Vivado 2025.2.

multiplier.v

RTL code:

```
module multiplier_8bit(  
    input clk,  
    input [7:0] A,  
    input [7:0] B,  
    output reg [15:0] P  
);  
always @(posedge clk)  
begin  
    P <= A * B;  
end  
  
endmodule
```

multiplier_tb.v

Testbench code:

```
module multiplier_tb;
```

```
reg [7:0] A;
```

```
reg [7:0] B;
```

```
wire [15:0] P;
```

```
multiplier_8bit DUT (
```

```
    .A(A),
```

```
    .B(B),
```

```
    .P(P)
```

```
);
```

```
initial
```

```
begin
```

```
    A = 8'd10;
```

```
    B = 8'd5;
```

```
    #10;
```

```
    A = 8'd15;
```

```
    B = 8'd3;
```

```
    #10;
```

```
A = 8'd20;
```

```
B = 8'd4;
```

```
#10;
```

```
A = 8'd25;
```

```
B = 8'd2;
```

```
#10;
```

```
A = 8'd255;
```

```
B = 8'd2;
```

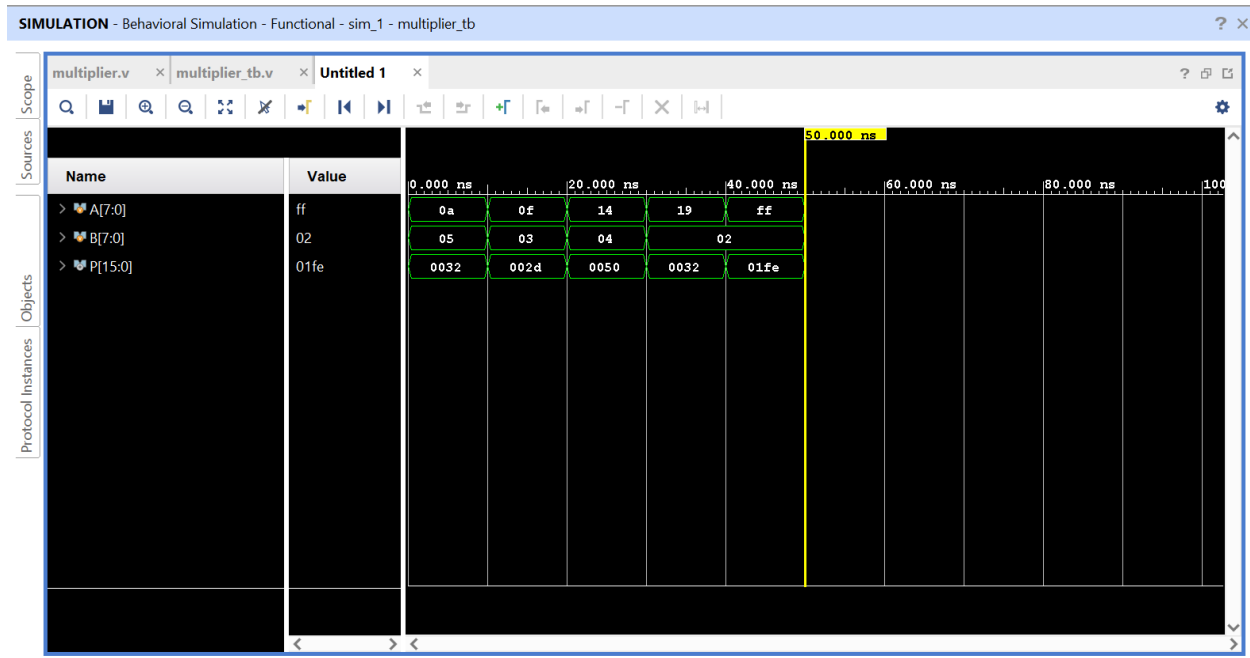
```
#10;
```

```
$finish;
```

```
end
```

```
endmodule
```

RTL Simulation Results:



Timing Constraints:

A timing constraint file (XDC) was created to define the clock.

Constraint Used:

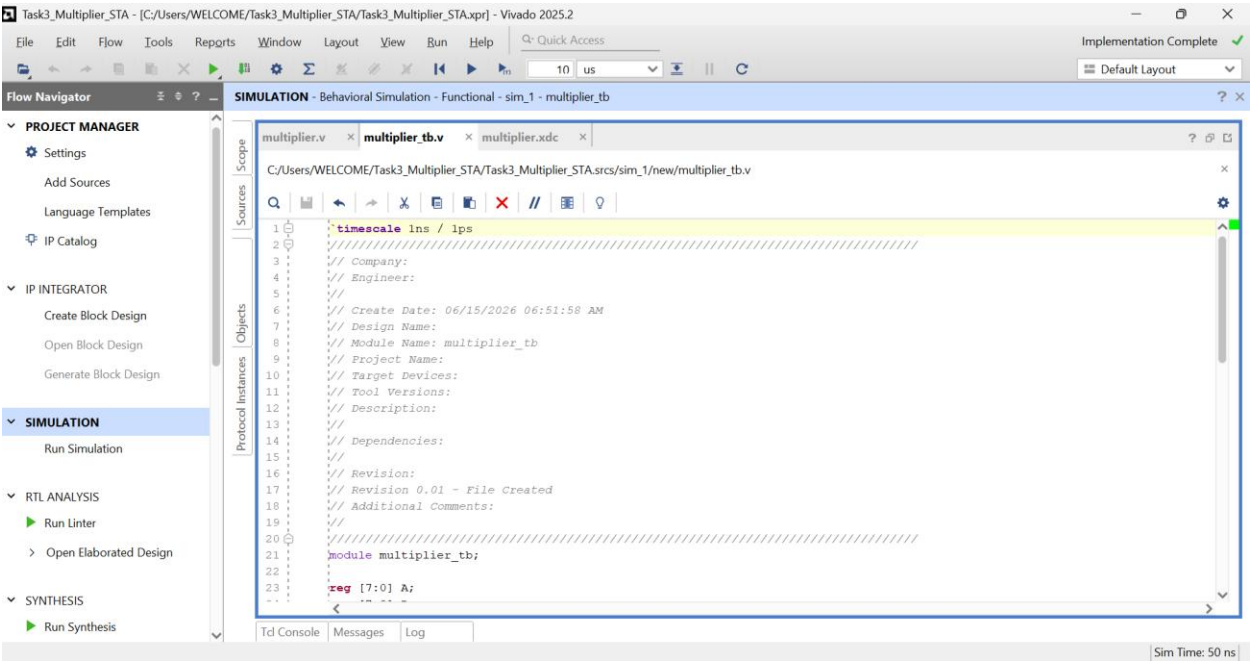
create_clock -period 10.000 [get_ports clk]

Clock Frequency:100MHZ

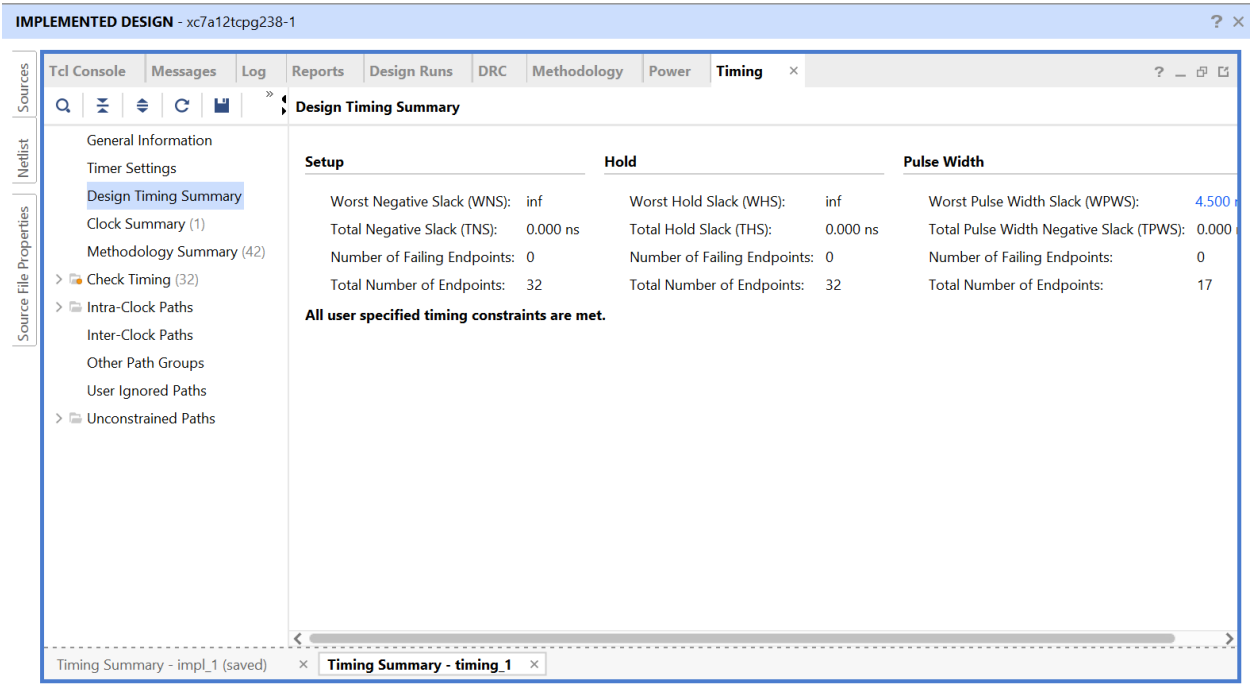
Procedure:

- RTL design was synthesized using Vivado.
- Implementation was performed.
- Timing reports were generated after implementation.

Synthesis Completed&Implementation Completed:



Static Timing Analysis Results:



Timing Closure Analysis

Observation:

- No setup violations were observed.
- No hold violations were observed.
- All user-specified timing constraints were met.
- Timing closure was successfully achieved.

Possible Fixes (if violations occur):

- Pipelining
- Logic optimization
- Register balancing
- Clock period relaxation

Conclusion:

Static Timing Analysis was successfully performed on the 8-bit multiplier design. The implemented design satisfied all timing constraints under a 10 ns clock period. No setup or hold violations were detected, and timing closure was successfully achieved.